

Event-Viewing

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

One of the employees at your company has their computer infected by malware! Turns out every time they try to switch on the computer, it shuts down right after they log in. The story given by the employee is as follows:

1. They installed software using an installer they downloaded online
2. They ran the installed software but it seemed to do nothing
3. Now every time they boot up and login to their computer, a black command prompt screen quickly opens and closes and their computer shuts down instantly.

See if you can find evidence for each of these events and retrieve the flag (split into 3 pieces) from the correct logs!

Hints

- Try to filter the logs with the right event ID
 - What could the software have done when it was ran that causes the shutdowns every time the system starts up?
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Tools Used

- `file` – Determine file type
 - `strings` – Extract printable character sequences
 - `exiftool` – Read metadata
 - `base64` – Decode Base64-encoded content
 - **Windows Event Viewer** – GUI tool for analyzing `.evtx` files
 - **PowerShell/Wevtutil** – Command-line Windows Event Log utility
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Complete Solution

Step 1: Download and Extract

Download the Windows Event Log file named `Windows_Logs.evtx` from the challenge page.

Step 2: Basic File Inspection

First, verify the file type:

```
file Windows_Logs.evtx
```

Expected output:

```
Windows_Logs.evtx: MS Windows 10-11 Event Log, version 3.2, 99 chunks (no. 98 in use), next record no. 8620
```

The file is a **Windows Event Log** (`.evtx`), containing system and security event records.

Step 3: Search for Visible Text

Check if the flag appears as plain text:

```
strings Windows_Logs.evtx | grep -i "picoCTF"
```

Output: (no results)

The flag isn't directly visible. We need to analyze specific event IDs.

Step 4: Examine Metadata

```
exiftool Windows_Logs.evtx
```

Output:

```
ExifTool Version Number      : 13.50
File Name                    : Windows_Logs.evtx
Directory                    : .
File Size                    : 7.4 MB
...
Error                        : Unknown file type
```

Nothing suspicious here.

Step 5: Understand the Scenario

The timeline of events is:

Step	Action	Likely Event ID
1	Software installation	1033 (MSI installer started) or 11724 (Package installed)
2	Software execution	4688 (Process creation)
3	Registry modification (persistence)	4657 (Registry value changed)
4	System shutdown	1074 (System shutdown initiated)

The hint suggests the malware added itself to startup, causing shutdowns every time the user logs in.

Step 6: Important Windows Event IDs

Event ID	Log	Description
1033	Application	Windows Installer started
1074	System	System shutdown/restart initiated
4657	Security	Registry value was modified

Event ID	Log	Description
4688	Security	A new process was created
7045	System	A service was installed
6005	System	Event log started (system boot)
6006	System	Event log stopped (system shutdown)
6008	System	Unexpected shutdown

For a comprehensive list, refer to:

- [Microsoft Learn Appendix L](#)
 - [Tenfold Security Event IDs](#)
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Step 7: Filter Events by ID

Using Windows Event Viewer or PowerShell, filter the logs for the relevant event IDs.

Event ID 1033 – Software Installation

This event confirms the user installed software. It may contain the first part of the flag.

Found data:

```
cGljb0NURntFdjNudF92aTN3djNyXw==
```

Event ID 4657 – Registry Modification

This event tracks changes to the Windows Registry. Malware often adds itself to `Run` or `RunOnce` keys for persistence. This log contains the second part of the flag.

Found data:

```
MXNfYV9wcjN0dH1fdXMzZnVsXw==
```

Event ID 1074 – System Shutdown

This event logs when the system is shut down or restarted. The malware likely initiated a shutdown after login. This log contains the third part of the flag.

Found data:

```
dDAwbF84MWJhM2ZlOX0=
```

Step 8: Combine the Flag Pieces

The three Base64 strings together form the complete flag:

```
cGljb0NURntFdjNudF92aTN3djNyXw==  
MXNfYV9wcjN0dH1fdXMzZnVsXw==  
dDAwbF84MWJhM2ZlOX0=
```

Save them to a file:

```
nano 64.txt
```

Paste the three Base64 strings (one per line).

Step 9: Decode Base64

```
cat 64.txt | base64 -d
```

Output:

```
picoCTF{<REDACTED>}
```

The three pieces concatenate to form the complete flag.

Step 10: Alternative – Decode Each Piece Separately

```
echo "cGljb0NURntFdjNudF92aTN3djNyXw==" | base64 -d  
echo "MXNfYV9wcjN0dH1fdXMzZnVsXw==" | base64 -d
```

```
echo "dDAwbF84MWJhM2ZlOX0=" | base64 -d
```

Then combine the outputs manually.

Step 11: Alternative – Python One-Liner

```
python
import base64
pieces = [
    "cGljb0NURntFdjNudF92aTN3djNyXw==",
    "MXNfYV9wcjN0dHlfdXMzZnVsXw==",
    "dDAwbF84MWJhM2ZlOX0="
]
flag = ''.join(base64.b64decode(p).decode() for p in pieces)
print(flag)
```

✅ Final Result

```
picoCTF{<REDACTED>}
```

Note: The actual flag is redacted here. In the real challenge, decoding the Base64 strings from the event logs will reveal the complete flag.

🧠 Key Concepts

Concept	Description
Windows Event Log (.evtx)	A binary file format that stores system, security, and application events.
Event ID	A unique identifier for each type of Windows event.
Event ID 1033	Windows Installer started – indicates software installation.

Concept	Description
Event ID 4657	Registry value modified – often used for persistence mechanisms.
Event ID 1074	System shutdown initiated by a process – indicates shutdown events.
Persistence	Malware techniques to ensure execution after reboot (Registry Run keys, Scheduled Tasks, Services).
Base64 Encoding	A common encoding used to hide flag pieces in logs.

Pro Tips

1. **Use wevtutil on Windows** – Command-line tool for querying `.evtx` files:

```
wevtutil ql /f:text Windows_Logs.evtx /q:"*[System[EventID=1074]]"
```

cmd

2. **Filter by Event ID** – In Windows Event Viewer, use `Filter Current Log` and enter the Event ID.
3. **Malware persistence locations** – Check:
 - `HKCU\Software\Microsoft\Windows\CurrentVersion\Run`
 - `HKLM\Software\Microsoft\Windows\CurrentVersion\Run`
 - `Task Scheduler`
4. **Registry Event ID 4657** – Requires `Audit Registry` to be enabled in Group Policy.
5. **Base64 often has = padding** – The presence of `=` at the end of a string is a strong indicator of Base64 encoding.
6. **Base64 of “picoCTF{” starts with cG1jb0NURnt** – Recognizing this pattern helps identify flag pieces quickly.

Alternative Tools

Tool	Purpose
Windows Event Viewer	GUI tool for browsing <code>.evtx</code> files
wevtutil	Command-line tool for querying event logs
PowerShell <code>Get-WinEvent</code>	<code>Get-WinEvent -Path .\Windows_Logs.evtx -FilterXPath "*[System[EventID=1074]]"</code>
EvtxCmd (libevtx)	Command-line tool for parsing <code>.evtx</code> files on Linux
Python-evtx	Python library for parsing <code>.evtx</code> files

References

- [Windows Event IDs Reference \(Tenfold Security\)](#).
- [Appendix L: Events to Monitor \(Microsoft Learn\)](#).
- [Event IDs \(SS64\)](#).